

Spencer Hadlock

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EDUCATION

Cornell University | Duffield College of Engineering

Ithaca, NY

Bachelor of Science in Electrical and Computer Engineering | GPA: 3.74

Expected: December 2026

George & Marie Zahakos Scholarship Recipient (2023-26)

State University of New York: Westchester Community College

Valhalla, NY

Associate of Science in Liberal Arts and Sciences: Mathematics and Science | GPA: 3.92

May 2024

Phi Theta Kappa Honor Society | Selected as 1 of 6 Valedictorian Candidates in a Class of 1,300

ENGINEERING EXPERIENCE

Cornell University Department of Electrical and Computer Engineering

Ithaca, NY

Course Development Intern

May 2026 – August 2026

- Developed extensive SystemVerilog integration testbenches for a single-cycle process with SPI and memory bus
- Utilize positive and negative testing to validate malformed or invalid inputs, reset behavior and timing-sensitive interactions,
- Diagnosed and isolated module failures using Surfer for detailed waveform analysis and cross-module signal tracing
- Built a submission workflow for 120 students to improve lab tracking, organization, and grading in a 'data-sheet' style

Nexus Project Team

Ithaca, NY

Electrical Subteam

September 2025 – Present

- Designed Altium 5 PCBs for autonomous robot's electrical systems across power, communication and signal processing
- Documented design notes, wiring references, and team procedures on local team wiki for current and future members
- Coordinated board deliverables for sensor, motor, and power needs of mechanical and software subteams for integration
- Supported PCB bring-up by reviewing schematics, validating connections, and debugging boards to pass design rule check

Lewisboro Electrical Services

South Salem, NY

Apprentice Electrician

June 2020 – Present

- Installed and upgraded residential and commercial electrical systems following NEC, drawings, and local building code
- Wired service installs, lighting circuits, receptacles, switches, and service equipment for renovation and new-build projects
- Troubleshoot issues on service calls and across active job sites, resolving integration with other trades and building stages
- Programmed Lutron HomeWorks and RadioRA 3 systems using Designer Pro for lighting, shades, and control automation

PROJECTS

Direct-Mapped Cache

RTL Design (SystemVerilog)

- Designed a direct-mapped cache in SystemVerilog with a fully combinational tag-comparison and hit-detection path to support single-cycle cache hits, paired with an FSM-controlled miss-handling and refill path for multi-cycle memory
- Implemented a wait-based memory-side handshake and developed directed SystemVerilog testbenches to verify cache hits, misses, refills, replacement behavior, and correct data delivery through the cache's external processor and memory interfaces

OFDM Audio Communication System

Digital Communications and Signal Processing

- Built an end-to-end OFDM audio communication system in Python, achieving 70 kbps reliable data transmission over an acoustic channel through QAM, OFDM subcarrier mapping, frame synchronization for real-time audio signals
- Implemented Gray-coded constellations, adaptive subcarrier bit allocation, frame synchronization, clock skew correction, and iterative channel equalization techniques to meaningfully improve overall system and signal transmission reliability

Multi-Label Digit Classification

Machine Learning Project

- Developed a dual-output image classification system in Python to accurately identify pairs of overlaid handwritten digits from labeled training data, requiring simultaneous prediction of both visible digit classes within each given input image
- Trained and benchmarked CNN, SVM, and several baseline classification models, applying preprocessing, data augmentation, cross-validation, and error analysis to reduce overfitting and reach approximately 97% overall test accuracy

ADDITIONAL

HDL & Digital Design Skills: SystemVerilog, RTL Design, Testbench Development, Waveform Debugging, Intel Quartus

Programming: Python, C, C++, Java, MATLAB, ARM Assembly, SQL, LaTeX

Hardware & Tools: Altium Designer, Surfer, Linux, Git, PCB Design, Embedded Systems, Verilator, IVerilog

Interests: Backpacking, Skiing, Rock Climbing, Homemade Yogurt, Mountain Biking, Slacklining Speed Cubing, Sudoku

Clubs: Cornell IEEE Committee Member, Climbing Club, Slackline Club, Cornell HFTC, and Club Lacrosse